

Yingrui Hou

Simulation & Data Scientist | Calibration, Validation & Modelling for Physical Systems

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SUMMARY

Simulation and data scientist with a background in experimental physics and several years of experience building C++/Python workflows for calibration, validation, inference, and monitoring on complex measurement systems.

My core strength is connecting simulations, noisy measurements, and statistical models to produce decision-ready results. I work best on problems where the true state is only indirectly observable and model credibility depends on calibration, bias correction, uncertainty-aware analysis, and rigorous validation, with strong intuition for instruments, detector behaviour, measurement limitations, and test-data interpretation.

CORE SKILLS

- Simulation and modelling: physical-system modelling, simulation-to-measurement alignment, Geant4, ROOT / RooFit
- Calibration and validation: model validation, calibration, bias correction, uncertainty-aware analysis
- Data science and inference: statistical modelling, anomaly detection, monitoring, dataset-shift handling, probabilistic inference
- Engineering workflow: Python, C/C++, Linux, workflow automation, reproducible analysis pipelines
- Experimental systems: noisy measurement systems, detector response, test-data interpretation, instrumentation-aware analysis

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher | CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration | 2021-2025

- Built C++/Python workflows to align simulations with noisy measurement data through calibration, validation, and model correction for decision-facing analyses.
- Developed statistical and simulation-based methods to estimate latent system behaviour when direct ground truth was unavailable.
- Built monitoring and diagnostic pipelines to compare expected versus observed behaviour, detect anomalies, and support root-cause analysis.
- Combined detector understanding, test data, and model outputs to improve calibration quality and reduce structured bias across operating conditions.
- Worked in Linux-based shared codebases with reproducible workflows and collaborative review discipline, translating validation results into calibration updates, software fixes, and improved model credibility.

Doctoral Research | University of Chinese Academy of Sciences | 2016-2021

- Developed Geant4/C++ simulation, signal-modelling, and calibration workflows during doctoral research to connect detector response, measurement data, and system-level performance studies.

SELECTED OUTCOMES

- Quantified that detector-response non-uniformity contributed less than 1% to system-resolution degradation in a detailed measurement-to-model study.
- Improved reconstructed-energy resolution by about 2x through time-resolved signal analysis and calibration on realistic measurement data.

EDUCATION	Joint Master and PhD in Particle Physics - University of Chinese Academy of Sciences, 2016-2021 BSc in Applied Physics - China University of Mining and Technology, 2010-2014
LANGUAGES	Chinese: Native speaker, English: C1, French: B1 to B2